

Epilepsy and Neuronal Excitability

Polygenic risk of generalized and focal epilepsy, febrile seizures, ionic channels (SCN1A) and pharmacogenetics of antiepileptics

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The loci load of generalized epilepsy is at percentile 9.3. The common loci PCDH7, VRK2/FANCL, STX1B and CHRM3 have no risk allele. The febrile seizures are at 24.2. In pharmacogenetics, there is no **HLA-A*31:01** (rs1061235 A/A), the marker of severe skin reactions with carbamazepine. CYP2C19 is normal (*1/*1: without *2 and without *17) and **EPHX1** has the usual form (Tyr113).

Unfavourable. The **SCN1A rs6732655** has 2 risk alleles (T/T). It is the strongest common locus of generalized epilepsy, but the effect is small ($\sim 1.1\times$ per copy). **ZEB2** has 2 alleles (A/A) for absence epilepsy and 2q24 (SCN1A/SCN2A) for febrile seizures has 1 allele. The CYP2C9 is *1/*3 (intermediate metabolizer), which matters to phenytoin, which accumulates. BDNF Val66Met is C/T (1 Met allele, without judgment; see §34).

What does it mean? The global polygenic risk of epilepsy is typical: genomic scores at the 69.5 percentiles (the same as §23), 54.5 and 24.2, and loci load of all epilepsies at 52.1. The most favourable signal is the generalized one; the focal one is in 66.5 (with only 8 loci). The data with practical consequence is pharmacogenetic: if ever an antiepileptic is needed, phenytoin requires a lower maintenance dose, and carbamazepine does not have the contraindication of HLA-A*31:01.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (statistical imputation, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 14 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 13 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 3 PGS Catalog scores calculated locally with 2,379,582 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 98 loci, 96 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more risk of epilepsy (all forms). The PGS004881 is the same as §23, with the same value.

Epilepsy

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Epilepsy (genomic)	PGS004881 (Jermy 2024, statistical method, 605 thousand variants)	599,669 605,432 (weighted coverage 99%)	69.5	TYPICAL	typical
Epilepsy (genomic)	PGS002760 (Mars 2022, Bayesian method, 836 thousand variants)	829,198 835,537 (weighted coverage 99%)	54.5	TYPICAL	typical
Epilepsy (genomic)	PGS018994 (Chen 2025, Bayesian method, 984 thousand variants)	950,715 983,823 (weighted coverage 96%)	24.2	TYPICAL	typical

- **Epilepsy (genomic):** In §23 (percentile 69.5).

4. Key variants by topic

Ionic channels and replicated loci of epilepsy (small effects, ~1.1–1.4 per copy) and the pharmacogenetic variants of antiepileptics with clinical orientation.

Sodium channels and excitability

SCN1A (Nav1.1) is the sodium channel of inhibitory interneurons; BDNF modulates plasticity and excitability.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SCN1A	rs6732655		T/T	chip	T (2 copy/s)	unfavourable	SCN1A region (sodium channel Nav1.1 of inhibitory interneurons): the strongest locus of generalized epilepsy [GWAS Catalog: Epilepsy, T allele, OR 1.12, p=9E-10, PMID 25087078]	Strong
SCN1A/ SCN2A (febrile seizures)	rs6432860		A/G	chip	G (1 copy/s)	unfavourable (light)	2q24 locus of febrile seizures [GWAS Catalog: Febrile seizures, G allele, OR 1.34, p=2E-16, PMID 25344690]	Moderate
BDNF	rs6265	Val66Met	C/T	chip	T (1 copy/s)	without judgment	Val66Met: T allele (Met) reduces activity-dependent BDNF release (see §34)	Weak

Common loci of epilepsy (GWAS)

Replicated loci of generalized epilepsy in major association studies.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PCDH7	rs28498976		G/G	chip	A (0 copy/s)	favourable	Protocadherin 7: generalized epilepsy [GWAS Catalog: Epilepsy, allele A, OR 1.11, p=5E-9, PMID 25087078]	Moderate
ZEB2	rs13020210		A/A	imputation DR2=0.97	A (2 copy/s)	unfavourable (light)	ZEB2 (Mowat-Wilson syndrome): generalized epilepsy; G allele is associated with less epilepsy from childhood absences, so A is the risk [GWAS Catalog: Childhood abscess epileptic, allele G with protective sense, p=2E-8, PMID 30531953]	Moderate
VRK2/ FANCL	rs2947349		A/A	chip	C (0 copy/s)	favourable	2p16.1: generalized epilepsy (shared with schizophrenia) [GWAS Catalog: Epilepsy, C allele, OR 1.23, p=1E-8, PMID 25087078]	Moderate
STX1B	rs1046276		C/C	chip	T (0 copy/s)	favourable	Syntaxin 1B (release of neurotransmitters): generalized epilepsy and febrile [GWAS Catalog: Juvenile myoclonic epileptic, allele T, p=3E-11, PMID 30531953]	Moderate
CHRM3	rs12059546		A/A	chip	G (0 copy/s)	favourable	Muscarinic receptor M3 [GWAS Catalog: Generalized epileptic, G allele, OR 1.42, p=4E-8, PMID 22949513]	Weak

Pharmacogenetics of antiepileptics

Enzymes that metabolize phenytoin, clobazam and carbamazepine, and HLA of severe skin reactions.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP2C9	rs1799853	*2	C/C	chip	T (0 copy/s)	favourable	*2: slower metabolism of phenytoin (lower dose)	Strong
CYP2C9	rs1057910	*3	A/C	chip	C (1 copy/s)	unfavourable (light)	*3: much slower metabolism of phenytoin; more toxicity and skin reactions	Strong
CYP2C19	rs4244285	*2	G/G	chip	A (0 copy/s)	favourable	*2: slow metabolizer of clobazam, brivaracetam and diazepam	Strong
CYP2C19	rs12248560	*17	C/C	chip	T (0 copy/s)	without judgment	*17: rapid metabolizer of clobazam and diazepam	Strong
HLA-A*31:01 (tag)	rs1061235	A*31:01	A/A	chip	T (0 copy/s)	favourable	The T allele marks HLA-A*31:01: syndrome DRESS and severe skin reactions with carbamazepine (and oxcarbazepine)	Strong
EPHX1	rs1051740		T/T	chip	C (0 copy/s)	without judgment	Y113H (C allele, His113): slower epoxide hydrolase; plus carbamazepine-10, 11-epoxide and lower doses	Weak

5. GWAS loci burden

Sum of loci GWAS Catalog ($p \leq 510-8$) by type of epilepsy. Alleles are oriented towards more risk and the sum is compared with the reference population. Focal epilepsy and febrile seizures have few loci.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Epilepsy (all)	17 (16)	15 of 32	16	52.1	typical
Generalized epilepsy	58 (57)	54 of 114	57	9.3	favourable
Focal epilepsy	8 (8)	6 of 16	8	66.5	typical
Febrile seizures	15 (15)	12 of 30	15	24.2	typical

6. Favourable

- **Generalized epilepsy:** loci load at percentile 9.3 (58 loci). **Moderate.**
- **Common loci without risk alleles:** PCDH7 G/G, VRK2/FANCL A/A, STX1B C/C and CHRM3 A/A. Moderate (small effects).
- **Febrile seizures:** loci load at 24.2. **Weak.**
- **HLA-A*31:01 absent** (rs1061235 A/A): without increased risk of DRESS and severe skin reactions with carbamazepine and oxcarbazepine. Strong (marker).
- **CYP2C19 *1/*1** (rs4244285 G/G, rs12248560 C/C): normal metabolism of clobazam, diazepam and brivaracetam. **Strong.**
- **EPHX1 Y113H T/T:** epoxide hydrolase (carbamazepine active metabolite). **Weak.**

7. Unfavourable

- **SCN1A rs6732655 T/T**: 2 risk alleles in the strongest locus of generalized epilepsy ($\sim 1.1\times$ per copy). Moderate (small effect, compensated by the rest of loci).
- **ZEB2 rs13020210 A/A**: 2 risk alleles of childhood absences (small effect; the sense comes from the protective G allele). **Weak**.
- **2q24 rs6432860 A/G**: 1 allele of febrile seizures ($\sim 1.3\times$). Weak (relevant especially in childhood).
- **CYP2C9 *1/*3** (rs1057910 A/C): intermediate metabolizer. Phenytoin accumulates and pharmacogenetic guidelines recommend a dose of maintenance $\sim 25\%$ lower, with serum levels. It also affects warfarin, NSAIDs and sulfonylureas (see profile). **Strong**.
- **Typical global risk**: scores in the percentiles 69.5, 54.5 and 24.2; focal epilepsy with loci at 66.5. **Moderate**.

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Intercellular Communication	Excitation-inhibition balance (SCN1A of GABAergic interneurons, STX1B of synaptic release); generalized epilepsy with loci at percentile 9.3	Favourable
Neuroinflammation	The neuroinflammation low the convulsive threshold with age; without specific risk genes here	Neutral
Sleep disturbance	Sleep deprivation is the most common trigger of seizures; protecting sleep protects the threshold	Conditional
Hematoencephalic Barrier dysfunction	The barrier breakdown (stroke, small vessels) is the main cause of late-onset epilepsy	Conditional
Vascular Aging	After the 60 years, most of the new epilepsys are vascular; LDL and ApoB low also protect by this route	Conditional
Mitochondrial dysfunction	Without mitochondrial mutations of myoclonic epilepsy tested (m.8344A>G absent, see haplogroup report)	Favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Pharmacogenetic record:** include in the clinical record **CYP2C9 *1/*3** (phenytoin: maintenance dose ~25% lower and serum levels; also warfarin, NSAIDs and sulfonylureas) and HLA-A*31:01 negative. Suggestion (not applied).
2. **HLA-B*15:02 was not read.** If one day start carbamazepine, oxcarbazepine or phenytoin, the doctor decides whether to type HLA; the risk is low in this profile, but the test is cheap. Suggestion (not applied).
3. **Protect the convulsive threshold:** regular sleep of 7–9 hours, zero alcohol (already fulfills) and care with drugs that lower the threshold (high dose bupropion, tramadol, some quinolonic antibiotics), especially in combination. Suggestion (not applied).
4. **Late onset epilepsy:** the main prevention is vascular, and it is already in the plan (LDL and ApoB low, exercise, controlled blood glucose). A first episode of loss of consciousness, sudden confusion or involuntary movements justifies neurological evaluation, because it may be the first sign of cerebrovascular disease. Suggestion (not applied).
5. **Melatonin 0.3–1 mg** (already in the plan): stabilizes sleep, with no known proconvulsive effect at these doses. Suggestion (not applied).

10. Limitations

- Monogenic epilepsy (SCN1A, SCN2A, KCNQ2, DEPDC5 and others) depends on rare variants that the chip does not read. The polygenic risk explains a small part of the common epilepsy.
- Focal epilepsy and febrile seizures have few loci; they tend to.
- HLA-B*15:02 and other pharmacogenetic alleles HLA were not read. HLA-A*31:01 is inferred by a marker (rs1061235), with good but not perfect agreement.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). The absent SNPs come from local statistical imputation (public panel of complete genomes to 30x), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed. The alleles of HLA-A*31:01 and EPHX1 were verified in the literature.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (3 genomic scores of epilepsy). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510-8$), 4 sets (98 loci), alleles oriented by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** ILAE 2018 and 2023 (GWAS of epilepsy), Feenstra 2014 (febrile seizures), Jermy 2024, Mars 2022 and Chen 2025 (scores), Karnes 2021 (CPIC: CYP2C9, HLA-B and phenytoin), Phillips 2018 (CPIC: HLA and carbamazepine), McCormack 2011 (HLA-A*31:01), GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Sodium channels and excitability	SCN1A	rs6732655	T/T	chip	T	2	strand + (palindromic; assumed +)	2 risk allele copies
Sodium channels and excitability	SCN1A/SCN2A (febrile seizures)	rs6432860	A/G	chip	G	1	strand +	1 copy of risk allele
Sodium channels and excitability	BDNF	rs6265	C/T	chip	T	1	strand +	1 copy of effect allele
Common loci of epilepsy (GWAS)	PCDH7	rs28498976	G/G	chip	A	0	strand +	without the risk allele
Common loci of epilepsy (GWAS)	ZEB2	rs13020210	A/A	imputation DR2=0.97	A	2	strand +	A/A: 2 copies of allele A, the most risky in this locus
Common loci of epilepsy (GWAS)	VRK2/FANCL	rs2947349	A/A	chip	C	0	strand +	without the risk allele
Common loci of epilepsy (GWAS)	STX1B	rs1046276	C/C	chip	T	0	strand +	without the risk allele
Common loci of epilepsy (GWAS)	CHRM3	rs12059546	A/A	chip	G	0	strand +	without the risk allele
Pharmacogenetics of antiepileptics	CYP2C9	rs1799853	C/C	chip	T	0	strand +	without the risk allele
Pharmacogenetics of antiepileptics	CYP2C9	rs1057910	A/C	chip	C	1	strand +	1 copy of risk allele
Pharmacogenetics of antiepileptics	CYP2C19	rs4244285	G/G	chip	A	0	strand +	without the risk allele
Pharmacogenetics of antiepileptics	CYP2C19	rs12248560	C/C	chip	T	0	strand +	0 copy of the effect allele
Pharmacogenetics of antiepileptics	HLA-A*31:01 (tag)	rs1061235	A/A	chip	T	0	strand + (palindromic; assumed +)	A/A: without the T allele that marks HLA-A*31:01
Pharmacogenetics of antiepileptics	EPHX1	rs1051740	T/T	chip	C	0	without data	T/T: Tyr113, epoxide hydrolase with usual activity